What the Model Learns from Coded Audio: Null-Calibrated Feature Attribution in Explainable Speech Emotion Recognition

Akshath R

Abstract—Speech emotion recognition is validated on uncompressed audio and deployed on coded audio. Whether that gap matters has been asked of accuracy; we ask it of the evidence, using models whose evidence can be read exactly. Fifteen classifiers and a small neural network are fitted to 436 named acoustic descriptors over CREMA-D, rendered in four conditions from one canonical reference: uncompressed WAV, MP3 and MP4/AAC at **64** kbit s^{-1}, and an AAC re-wrap that holds the codec output fixed while changing the container. Three results follow. First, the catastrophic cross-format collapse is train/serve mismatch and nothing else: a kernel machine trained on WAV scores 0.203 balanced accuracy on MP3, but the same model trained on MP3 scores 0.589 against 0.586 on clean audio, so the affective information survives the codec intact. Second, what the model learns from that information does change, and the change is legible because the features are named: an entropy tree fitted per condition selects the same root descriptor (prosody_intensity_std) in all four conditions with information gain stable to 0.003 bits, but AAC displaces the threshold by 1.3 % and replaces the depth-1 descriptor, while the MP3 tree does not. Third — and this is the methodological contribution — a codec effect on a feature-importance ranking is uninterpretable without a null, because the ranking is fragile. Refitting on identical audio with different tie-breaking retains 24.2 of 25 top features, but an informationally empty container re-wrap already costs 8.4 of them. Against that floor the codec effects are real but far smaller than they look: MP3 retains 12.6 and AAC 10.7 ($p \leq 1.1 \times 10^{-3}$, paired within resampled training sets), so roughly three fifths of the apparent displacement is generic perturbation sensitivity, not codec-specific. We report the complete 436-column inventory and the full per-condition split structure, and argue that attribution-shift claims in affective computing require a matched empty-perturbation control as standard.

Index Terms—Speech emotion recognition, explainable artificial intelligence, perceptual audio coding, feature attribution, decision trees, information gain, covariate shift, null calibration, affective computing.

I. Introduction

AUDIO almost never reaches an inference endpoint in the representation in which it was captured. Contact-centre archives hold MP3; streaming clients negotiate Opus or AAC; broadcast pipelines resample and requantise. Evaluation practice runs the other way: benchmarks are distributed as uncompressed WAV and read directly from disk.

The gap is usually dismissed on a reasonable argument. Any ingestion path decodes to linear PCM before feature extraction, so the delivery format is an implementation detail. Where the argument has been tested, it has been tested on accuracy, and the answers are mixed: Reddy and Vijayarajan [1] report codec effects on speech emotion recognition (SER) that are real, modest, and dependent on the front end.

Accuracy is the wrong instrument for a system that is increasingly required to say not only what it concluded but why. A deployed affective system may surface an attribution over acoustic evidence to a clinician, an auditor, or a quality-assurance analyst. That second output has its own robustness properties, and they are not implied by the first. Ghorbani et al. [2] established for image classifiers that perceptually indistinguishable inputs receiving identical predictions can be assigned materially different attributions; Alvarez-Melis and Jaakkola [3] showed the same fragility more generally. Both used adversarially constructed perturbations. A transcode is not adversarial. It is routine, ubiquitous, and nobody logs it.

A. Why interpretable models, and why named features

Answering an evidential question requires a model whose evidence can be inspected, and inspectability is not one property. An end-to-end audio model learns its own representation, so an attribution over one of its latent dimensions is not a claim about anything an acoustician would recognise. A model fitted to named descriptors in the GeMAPS [4] and openSMILE [5] tradition is different: an attribution over prosody_cpps is a claim about breathiness, and a decision-tree split on spectral_contrast_6_mean is a claim about the 6 kHz to 8 kHz band. This paper is confined to that regime deliberately. Everything reported here is exact rather than estimated, checkable against an acoustic reading, and reproducible from the released artefacts.

B. The problem with attribution-shift claims

A claim of the form “compression changed which features the model relies on” is easy to produce and hard to justify. Feature-importance rankings are notoriously unstable: two descriptors that split almost equally well

Manuscript prepared August 23, 2026.

A. R is an independent researcher (e-mail: akshath.r333@gmail.com).

Code, stimulus manifests, the feature table schema, and every result file underlying the tables and figures are released with this paper; see the Data Availability Statement.

can swap, and one swap near a tree's root reroutes every descendant. A ranking can therefore move a long way without the data having moved at all. The literature already contains the general warning — Adebayo et al. [6] show that some saliency methods are independent of the model they claim to explain, and Krishna et al. [7] show that different methods disagree on the same fitted model — but the specific discipline it implies, measuring the shift induced by a manipulation that carries no information, is not standard practice.

We adopt it here, and it changes the answer. Our design supplies a natural empty perturbation: the same codec output routed through two containers. Once that floor is measured, most of the apparent codec effect on importance rankings turns out to be generic sensitivity to any input perturbation.

C. Contributions

1) A matched-format decomposition of the codec effect. Training and serving in the same coded condition recovers essentially all of a catastrophic cross-format loss — up to 38.6 balanced-accuracy points — establishing that $64\,\mathrm{kbit\,s^{-1}}$ coding does not remove the affective information, it moves it out of the range the model was fitted on (Section VI-C).
2) Per-condition tree structure as a readable attribution. An entropy tree fitted independently on each condition exposes exactly which test the codec changes and which it leaves alone, at the level of a named descriptor and a numeric threshold (Section VI-D).
3) Null calibration for feature-importance shift. Three floors — estimator tie-breaking, actor resampling, and an informationally empty container re-wrap — with a paired within-sample design that separates the codec effect from all three (Section VI-E).
4) A diagnosable and cheap repair. Neutralising a single named descriptor recovers 98 % of a kernel machine's loss and 92 % of a network's, at no cost on uncompressed audio (Section VI-H).
5) Two negative results retained rather than smoothed. Post-hoc attribution methods disagree substantially on the same fitted model, and a global attribution ranking can report "nothing changed" while the classifier has stopped working (Sections VI-F2, VI-G1).
6) A complete, released inventory. All 436 descriptors are defined and enumerated (Appendix A), and every internal node of every per-condition tree is tabulated (Appendix B).

II. Related Work

A. Perceptual coding and speech task degradation

Degradation of speech systems under lossy coding is documented but shallowly characterised. Reddy and Vijayarajan [1] find that standard codecs reduce SER accuracy with a magnitude conditional on both codec family and feature representation, which already implies that coding interacts with the front end rather than degrading performance uniformly.

Of greater methodological consequence is a result from neural codec benchmarking. Wu et al. [8] demonstrate that signal-domain fidelity metrics fail to predict the retention of task-relevant information: a codec can score well under reconstruction error while discarding paralinguistic content that downstream tasks require. We take this seriously enough to report signal-domain displacement and model behaviour as separate quantities throughout, treating neither as a surrogate for the other.

B. Interpretability for tabular and audio models

Post-hoc attribution has converged on a small set of workhorses: local surrogate fitting [9], Shapley-value attribution [10] with an exact polynomial-time algorithm for tree ensembles [11], and permutation importance. For differentiable models, path-integral methods [12] and reference-based decompositions [13] are standard, and are available in consolidated form [14]. Broad surveys map the space [15]; Rudin [16] argues that for high-stakes decisions the post-hoc layer should be replaced by an intrinsically interpretable model, a position our Section VI-A result speaks to directly and unfavourably.

Transfer of these methods to audio is contested. Haunschmid et al. [17] argue that audio explanations should satisfy a listenability criterion and that transplanting image-segmentation logic onto time–frequency representations yields components with no perceptual correlate. Related formulations appear in [18] and [19], and Nasr et al. [20] extend the critique to speech emotion specifically, observing that a salient time–frequency region is not thereby an acoustically meaningful one. Attributing over named acoustic descriptors sidesteps this objection rather than answering it: every unit of attribution here has a phonetic reading by construction, at the cost of confining the model to hand-designed features.

C. Attribution stability as a validity requirement

The literature most directly load-bearing for this paper concerns the fragility of attribution itself. Ghorbani et al. [2] show that attributions can be substantially altered by perturbations below the threshold of perception while predictions and confidences remain stable. Adebayo et al. [6] demonstrate that certain widely deployed saliency methods are statistically independent of both model parameters and data-generating process, producing outputs with the surface form of explanation and none of its content. Krishna et al. [7] document that different explanation methods routinely disagree on the same model, and that practitioners resolve the disagreement by unprincipled means.

The joint implication is that attribution stability must be measured against a null, not presumed. Section VI-E operationalises that requirement for the specific object at issue here, a feature-importance ranking, and finds that the requirement is not a formality: it changes the size of the reported effect by a factor of roughly two and a half.

D. Distribution shift

The mechanism we identify is textbook covariate shift [21], with one unusual property. The shifted variable is not a proxy for anything the label depends on — masking it costs nothing on clean audio (Section VI-H) — so the shift is pure contamination rather than a change in the predictive relationship. That is what makes the repair as cheap as it turns out to be.

E. Where the gap lies

Coding effects are characterised for accuracy but not for evidence. Attribution fragility is characterised for vision under adversarial perturbation but not for audio under ecologically routine degradation. We identify no prior work that isolates the container under a verified signal-invariance constraint, and none that calibrates a feature-importance shift against a matched empty perturbation.

III. Corpus, Stimuli, and Conditions

A. Corpus

CREMA-D [22] comprises 7,442 clips from 91 actors, each delivering one of twelve fixed sentences in one of six acted emotions (anger, disgust, fear, happiness, neutral, sadness), with crowd ratings collected separately for voice-only, face-only and audio-visual presentation.

The fixed lexical inventory is methodologically load-bearing. Holding the words constant across emotions and conditions removes the lexical channel as a source of discriminative variance, so any recovered signal must be carried by acoustic realisation — which is precisely what perceptual coding operates on.

For reference throughout: CREMA-D's own crowd raters, hearing audio alone, matched the actor's intended emotion on only 45.5 % of clips, with per-class recall ranging from 0.966 (neutral) to 0.164 (sad). Models trained on the intended label can exceed this because they exploit systematic acting cues human listeners do not consciously decode; the figure is reported as context for absolute accuracy, not as a target.

B. Reference construction and conditions

The canonical reference is 16 kHz monophonic 16-bit linear PCM, loudness-normalised to EBU R128 ($I = -23$ LUFS, LRA $= 7$, TP $= -2$ dB). Coded conditions are produced from that reference by a single encoder invocation each, and all decoding passes through one identical FFmpeg call so that the decoder never confounds the codec.

The fourth condition is the identification device. roundtrip_wav is the AAC bitstream of mp4_aac64 decoded and re-wrapped as WAV: the same codec output, a different container and one additional int16 quantisation. It is not bit-identical to the direct MP4 decode — the WAV path requantises — but its median standardised feature difference is $|\mathrm{SMD}| = 7.6 \times 10^{-5}$, with only 12 of 436 descriptors exceeding 0.05. That is the noise floor of this study, and Section VI-E uses it as an informationally empty perturbation rather than merely reporting it as small.

Extraction over $7{,}442 \times 4 = 29{,}768$ files failed on exactly one clip in all four conditions, 1076_MTI_SAD_XX, which the corpus documentation lists as having no audio. The remaining 7,441 clips form the analysis set.

C. What the codec does to the descriptors

Fig. 2 characterises the manipulation before any model is involved. Two mechanisms are distinguishable and they behave differently.

MP3 abandons the top octave. Perceptual bit allocation quantises the highest band coarsely and zeroes masked bins, so spectral_contrast_6_mean moves from 16.38 to 46.54 and spectral flatness collapses in parallel (mean $0.0189 \rightarrow 0.0066$) because the residual spectrum is far less noise-like. Measured in training-set standard deviations the shift is 38.6σ: the descriptor arrives at inference time in a range no model fitted on clean audio has ever seen.

AAC shifts frame alignment. Encoder priming lengthens every clip by 32.68 ms, uniformly, which perturbs every frame-aligned temporal derivative a little rather than destroying one band. The largest AAC shifts are consequently all delta-MFCC means (mfcc_d1_01_mean, SMD $= -4.16$). We flag a caveat here rather than in the limitations, because it bears on how the number should be read: delta means sit near zero with small standard deviations, so their standardised shifts are inflated relative to their absolute change. The duration measurement is the harder evidence for the padding mechanism.

IV. Feature Representation

Each file is reduced to 436 named descriptors in four families, every frame-level contour collapsed to eight order statistics (mean, standard deviation, minimum, maximum, median, interquartile range, skewness, excess kurtosis). Appendix A defines every descriptor and enumerates every column; the summary is:

- spectral (114): centroid, bandwidth, roll-off (85/95 %), flatness, flux, entropy, slope, zero-crossing rate, RMS, seven-band spectral contrast, eight band-energy ratios, and explicit high-frequency survival ratios above 4 and 6 kHz;
- cepstral (240): 20 MFCCs with Δ and $\Delta\Delta$;
- prosodic (56), computed in Praat via Parselmouth [23]: F_0 statistics and slope, jitter (local, RAP, PPQ5, DDP), shimmer (local, dB, APQ3, APQ5, APQ11, DDA), HNR, formants F_1–F_5 with bandwidths, intensity, voiced-segment timing, and smoothed cepstral peak prominence (CPPS);
- chroma (24) and two global descriptors (duration, peak amplitude).

Spectral and cepstral descriptors are computed with librosa [24] at 512 point FFT, 400 sample window, 160 sample hop. Chroma tuning is pinned to zero rather than estimated, because an estimated tuning could itself

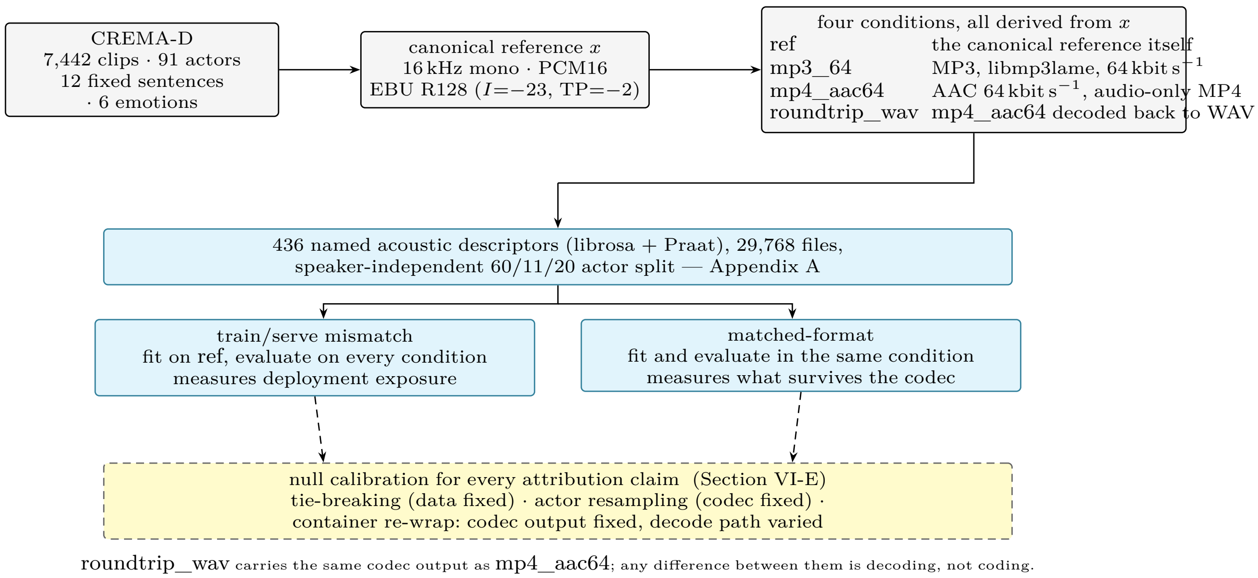

Fig. 1. The design. Every condition descends from a single canonical reference x, so no format contrast can be confounded by a divergent resampling or gain path. The same models are fitted twice — once on uncompressed audio and served compressed, once trained and served in the same condition — and the difference between those two readings is the paper's first result. Every attribution claim is evaluated against a floor measured on a manipulation that carries no coding information.

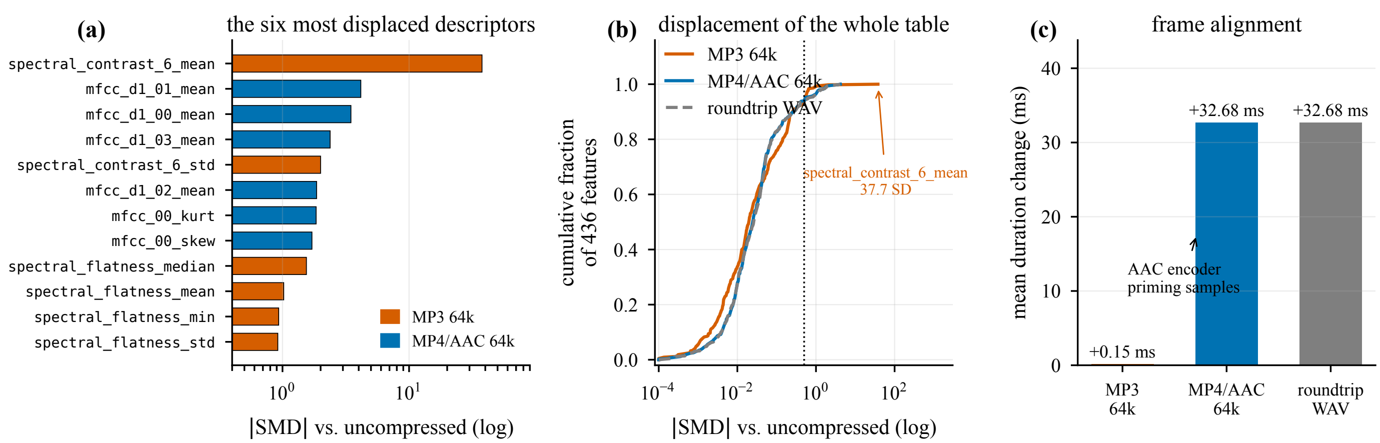

Fig. 2. The coding manipulation, measured on the feature table over 7,441 clips. (a) The six most displaced descriptors under each codec. MP3's displacement is concentrated: spectral_contrast_6_mean, which summarises the 6 kHz to 8 kHz band, moves by 37.7 pooled standard deviations. AAC's is diffuse and lands on delta-MFCC means, the frame-aligned temporal derivatives. (b) Empirical CDF of the absolute standardised mean difference of all 436 descriptors. The median shift is negligible (≈ 0.02); the tail is not. (c) AAC inserts encoder priming samples, lengthening every clip by 32.68 ms; MP3 adds 0.15 ms.

shift under compression and would silently become part of the effect being measured.

One descriptor is defined here because it recurs in the results and because its formula is the same one the trees of Section VI-D use for a different purpose. Per-frame spectral entropy is the Shannon entropy of the power spectrum normalised by its own bin count,

$$H_t = -\frac{1}{\ln K}\sum_{k=1}^{K} P_{k,t} \ln P_{k,t}, \qquad P_{k,t} = \frac{S_{k,t}}{\sum_{j=1}^{K} S_{j,t}}, \tag{1}$$

where $S_{k,t}$ is the power in bin k of frame t and K is the number of bins. Normalisation puts $H_t \in [0,1]$ and makes it scale-free: $H_t \to 1$ for a frame whose energy is spread evenly across frequency and $H_t \to 0$ for a pure tone. It falls when an encoder zeroes masked bins, which is why it moves under MP3 before the MFCCs do.

A. Splitting

Actors are partitioned 60/11/20 into train, validation and test with sex balance preserved, yielding 4,905/896/1,640 clips. Speaker-independent splitting is not a nicety here. A variance decomposition over the feature table gives mean $\eta^2 = 0.178$ for speaker against 0.098 for emotion, with 340 of 436 descriptors speaker-dominant; a random row split would leak voice identity into the test set and inflate every score reported below. Item alignment is enforced: the same clips appear in every condition, so a cross-condition comparison differs only in coding.

B. Separability

397 of 436 descriptors separate emotion at Bonferroni-corrected significance. The strongest single descriptors are mfcc_d1_00_std ($F = 1194$), mfcc_00_std ($F = 1159$) and prosody_intensity_std ($F = 1078$) — all variability measures, not mean levels. Emotion in acted speech lives in the dynamics, a fact that recurs in every attribution result below. By mutual information the ranking of families inverts the ranking by column count: chroma (0.098) > prosody (0.087) > spectral (0.082) > MFCC (0.061). MFCCs win on count, not on information per column. Fifty-one principal components are needed for 95 % of the variance, so the space is genuinely high-dimensional rather than a few latent factors wearing 436 names.

V. Models and the Splitting Criterion

A. The zoo

Fourteen scikit-learn classifiers spanning tree ensembles [25], [26], kernel machines [27], linear and discriminant models, and instance-based and probabilistic baselines [28], plus a PyTorch MLP (512-256-128 with batch normalisation and dropout 0.3, AdamW with a cosine schedule, class-weighted cross-entropy with label smoothing, early stopping on validation balanced accuracy) — fifteen trained models in all.

B. Entropy and information gain

Trees are grown with the Shannon-entropy criterion rather than Gini, so that the split rule reported in Section VI-D is the one these equations describe. For a node S carrying class weights w_c for each of the six emotions, with $w = \sum_c w_c$, the node entropy in bits is

$$H(S) \;=\; -\sum_{c=1}^{6} p_c \log_2 p_c, \qquad p_c \;=\; \frac{w_c}{w}. \tag{2}$$

A candidate test on descriptor f at threshold τ partitions S into $S_L = \{x : x_f \le \tau\}$ and S_R, and is scored by the information gain

$$\mathrm{IG}(S, f, \tau) \;=\; H(S) \;-\; \frac{w_L}{w} H(S_L) \;-\; \frac{w_R}{w} H(S_R). \tag{3}$$

The tree selects the (f, τ) maximising Eq. 3 at every node. Class weights are balanced, so w_c is the inverse class frequency rather than a raw count; CREMA-D is nearly balanced (1,271 clips for each of five emotions, 1,087 neutral), so the weights are all close to unity.

C. A worked example

Because the weights equalise the classes exactly, the root entropy of every tree in this paper is

$$H(S_{\mathrm{root}}) \;=\; -\sum_{c=1}^{6} \tfrac{1}{6} \log_2 \tfrac{1}{6} \;=\; \log_2 6 \;=\; 2.5850 bits. \tag{4}$$

On uncompressed audio the maximising test is prosody_intensity_std ≤ 8.2184, which sends $w_L = 3098.67$ of $w = 4905.00$ weighted samples left and $w_R = 1806.33$ right, with $H(S_L) = 2.3849$ and $H(S_R) = 2.0891$ bits. Substituting into Eq. 3,

$$\begin{aligned}\mathrm{IG} &= 2.5850 - \tfrac{3098.67}{4905.00}(2.3849) - \tfrac{1806.33}{4905.00}(2.0891) \\ &= 2.5850 - 1.5066 - 0.7693 \;=\; 0.3090 bits.\end{aligned} \tag{5}$$

So the single most informative question about a CREMA-D clip — how much does its loudness vary? — resolves 12 % of the total label uncertainty. Every threshold, entropy and gain reported below is computed this way, and the complete node tables are in Appendix B.

TABLE I
Trained and tested on uncompressed audio, 20 held-out speakers, six-way. Chance $= 0.167$; human voice-only $= 0.455$.

Model	Acc.	Balanced acc.	Macro F_1	κ
ANN (PyTorch)	0.599	0.599	0.596	0.518
LightGBM	0.590	0.591	0.584	0.508
SVM (RBF)	0.585	0.586	0.581	0.502
LDA (shrinkage)	0.585	0.585	0.580	0.502
Linear SVM	0.583	0.584	0.577	0.500
XGBoost	0.581	0.582	0.575	0.497
MLP (scikit-learn)	0.573	0.574	0.569	0.488
HistGradientBoosting	0.573	0.573	0.566	0.487
Logistic regression	0.565	0.567	0.561	0.478
Random forest	0.532	0.535	0.517	0.439
AdaBoost	0.520	0.521	0.514	0.423
Extra trees	0.516	0.519	0.496	0.420
kNN ($k = 15$)	0.477	0.479	0.466	0.373
Gaussian NB	0.424	0.430	0.379	0.312
Decision tree	0.390	0.391	0.382	0.268
Dummy (majority)	0.171	0.167	0.049	0.000

VI. Results

A. In-format performance

Table I gives the uncompressed leaderboard. Two points matter for what follows, and neither is about the winner.

First, a single decision tree — the one model a human can read end to end — reaches 0.391 balanced accuracy against 0.599 for the network. Intrinsic interpretability costs a third of the achievable performance on this task. That is an argument against Rudin's prescription [16] in this specific setting, and in favour of post-hoc attribution over a strong model. Section VI-F2 then shows what that trade buys and what it costs.

Second, the spread between the top eight models is under three points, so the leaderboard ordering carries little information. The differences that matter in this paper are 10 to 38 points and appear only when the coding condition changes.

B. Train/serve mismatch

Table II and Fig. 3 report what happens when a model fitted on clean audio is served coded audio. The result is categorical rather than graded. Every tree-based model sits within ± 0.014 of its clean score on MP3. Every model that consumes descriptor values as continuous magnitudes loses between 12 and 38 points.

TABLE II
All models trained on uncompressed audio and served compressed. Balanced accuracy on 20 held-out speakers. Models are grouped by how they consume a descriptor value. roundtrip is AAC output re-wrapped as WAV — the container control, whose difference from mp4_aac64 has a median of 0.003 and a maximum of 0.010 (kNN) across the fifteen models.

	Model	ref	mp3_64	Δ MP3	mp4_aac64	Δ AAC	roundtrip
axis-aligned partitioning	LightGBM	0.591	0.586	−0.005	0.559	−0.032	0.562
	XGBoost	0.582	0.586	+0.004	0.560	−0.022	0.560
	HistGradientBoosting	0.573	0.559	−0.014	0.530	−0.043	0.533
	Random forest	0.535	0.532	−0.003	0.522	−0.013	0.522
	AdaBoost	0.521	0.511	−0.010	0.487	−0.034	0.484
	Extra trees	0.519	0.521	+0.002	0.512	−0.007	0.507
	Decision tree	0.391	0.382	−0.009	0.367	−0.023	0.371
magnitude-sensitive	kNN	0.479	0.362	−0.118	0.467	−0.012	0.457
	Gaussian NB	0.430	0.277	−0.152	0.437	+0.008	0.433
	Linear SVM	0.584	0.405	−0.179	0.471	−0.113	0.462
	MLP (scikit-learn)	0.574	0.383	−0.191	0.491	−0.083	0.494
	ANN (PyTorch)	0.599	0.354	**−0.246**	0.513	−0.087	0.512
	LDA	0.585	0.338	−0.247	0.507	−0.079	0.502
	Logistic regression	0.567	0.233	−0.334	0.472	−0.095	0.470
	SVM (RBF)	0.586	0.203	**−0.383**	0.499	−0.087	0.497

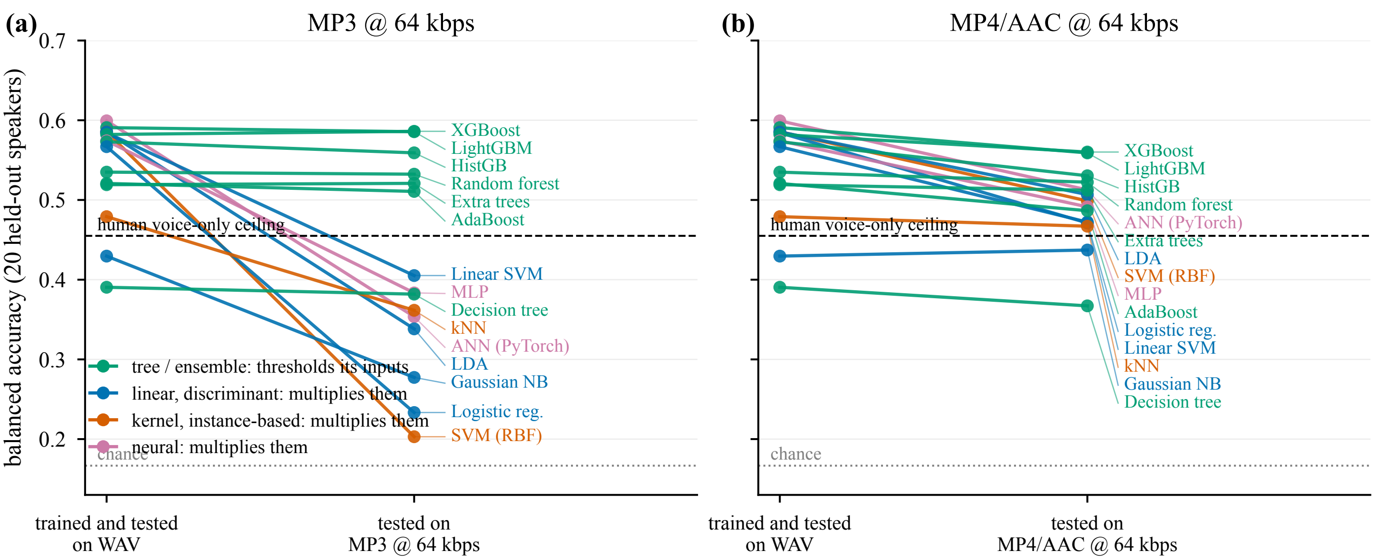

Fig. 3. Robustness to lossy coding under train/serve mismatch is determined by how a model consumes its inputs, not by how accurate it is. (a) Under MP3 at 64 kbit s^{-1} the tree family is flat while the magnitude-sensitive family collapses through the human ceiling toward chance. (b) Under MP4/AAC the same models degrade broadly and mildly, consistent with a diffuse frame-alignment perturbation rather than a single-band blowout. All models were trained on uncompressed audio only.

The mechanism is legible because the descriptors are named. A tree asks whether spectral_contrast_6_mean exceeds a threshold near 17; the answer is unchanged whether the descriptor reads 16.4 or 46.5. A linear or kernel model multiplies that 38σ excursion straight into its decision function.

We name the two groups by the property that separates them rather than by model family, because two members do not fit the obvious label. kNN computes distances and Gaussian NB evaluates per-descriptor likelihoods; neither multiplies a descriptor by a learned weight. What all eight share is that a descriptor's magnitude enters the decision continuously, whereas the seven tree-based models only ever ask on which side of a threshold it falls.

MP4/AAC behaves differently, costing everyone something and nobody very much (0.7 to 11 points regardless of family), consistent with the diffuse padding mechanism of Fig. 2(c). Container routing has no material effect: across the fifteen models the median difference between roundtrip and mp4_aac64 is 0.003 balanced accuracy and the maximum is 0.010, against codec effects one to two orders of magnitude larger.

C. Matched-format training

The mismatch result is a statement about deployment, not about the codec. To separate them, every model was refitted within each condition — same actors, same items, same hyper-parameters, coded audio in both training and test. Table III and Fig. 4 give the outcome, and it is unambiguous.

The RBF-SVM scores 0.589 on MP3 when trained on MP3, against 0.586 on clean audio — a recovery of

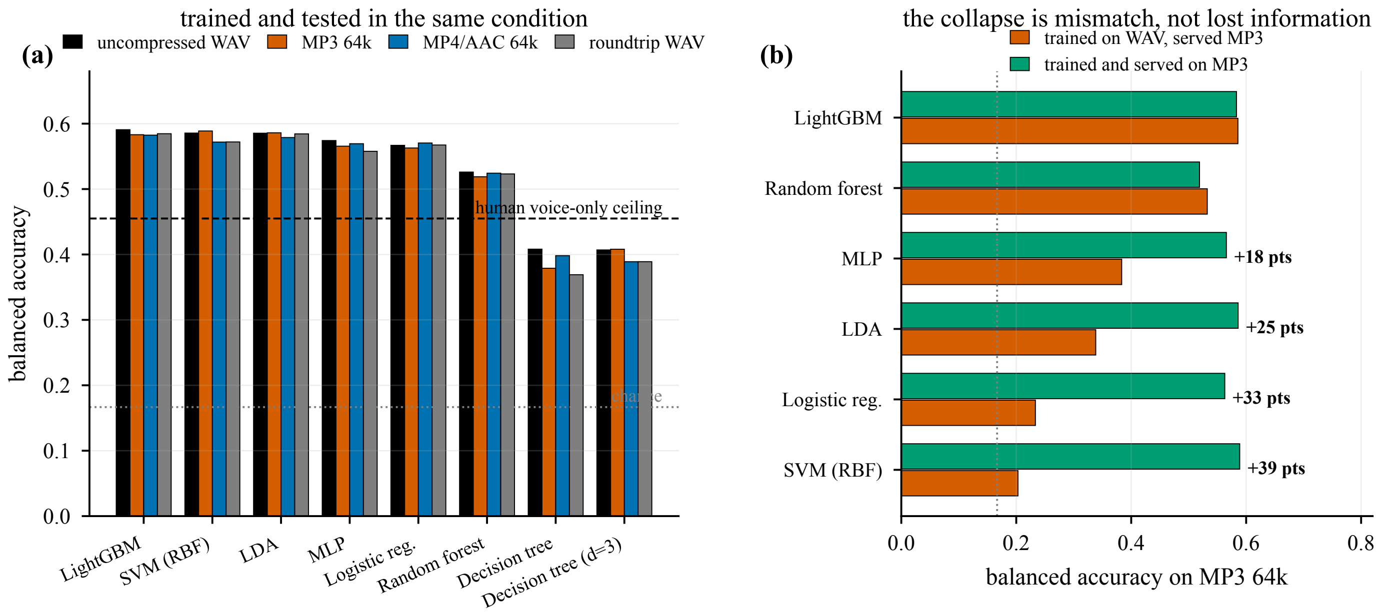

Fig. 4. Matched-format training removes the damage that mismatch creates. (a) Every model, fitted and evaluated in the same condition, sits within 0.008 to 0.039 balanced accuracy of its uncompressed score — including the two that lose 33 and 38 points under mismatch. (b) The same comparison on MP3 only. The affective information survives 64 $\mathrm{kbit\,s^{-1}}$ coding; what fails is the mismatch between the distribution a model was fitted on and the one it is served.

TABLE III
Matched-format training: balanced accuracy when a model is fitted and evaluated in the same condition. The rightmost column is the spread across all four conditions. Contrast with Table II, where the same models lose up to 38 points.

Model	ref	mp3_64	mp4_aac64	roundtrip	range
LightGBM	0.591	0.583	0.582	0.585	0.008
SVM (RBF)	0.586	0.589	0.572	0.572	0.017
LDA (shrinkage)	0.585	0.586	0.579	0.584	0.007
MLP (scikit-learn)	0.574	0.566	0.569	0.558	0.017
Logistic regression	0.567	0.563	0.570	0.567	0.008
Random forest	0.526	0.519	0.524	0.523	0.007
Decision tree ($d = 12$)	0.408	0.379	0.398	0.369	0.039
Decision tree ($d = 3$)	0.407	0.408	0.389	0.389	0.019

TABLE IV
Root and depth-1 splits by condition, from the depth-3 entropy tree. H and IG are in bits (Eqs. 2–3). L is the branch taken when the root test is satisfied.

Condition	Descriptor	Threshold	IG
root $H = 2.5850 bits = \log_2 6$			
ref	prosody_intensity_std	8.2184	0.3090
mp3_64	prosody_intensity_std	8.2210	0.3090
mp4_aac64	prosody_intensity_std	8.3241	0.3064
roundtrip_wav	prosody_intensity_std	8.3241	0.3064
depth 1, branch L			
ref	mfcc_d1_00_std	3.9965	0.1415
mp3_64	mfcc_d1_00_std	4.0175	0.1398
mp4_aac64	mfcc_00_iqr	60.5853	0.1408
roundtrip_wav	mfcc_00_iqr	60.5802	0.1408

38.6 balanced-accuracy points over its mismatched score of 0.203. Logistic regression recovers 33.0 points, LDA 24.8, the shallow MLP 18.2. No model's spread across the four conditions exceeds 0.039, and the two largest spreads belong to the single decision tree, whose variance is high for reasons Section VI-E makes precise.

Three consequences follow.

The codec does not remove the affective information. If 64 $\mathrm{kbit\,s^{-1}}$ MP3 destroyed the acoustic correlates of emotion, a model fitted on MP3 could not recover them. It does, to within three thousandths of the clean-audio score. The catastrophic column of Table II measures an artefact of deployment practice, not a property of the codec.

The tree family's advantage is an advantage under mismatch only. Under matched training, LightGBM's lead over the RBF-SVM is 0.008 on clean audio and −0.006 on MP3. The robustness ordering of Section VI-B is entirely a statement about out-of-range inputs.

The fix is format-specific, not general. Trained on mp4_aac64 and served MP3, the RBF-SVM falls to 0.167 — exactly chance. Matched-format training immunises against the codec it was trained on and no other, which is why Section VI-H's descriptor-level repair is the more practical remedy when the delivery format is not known in advance.

D. What the tree splits on, per condition

Accuracy parity does not entail evidential parity. Because the models are fitted to named descriptors, the question can be asked directly: given audio that supports the same accuracy, does the model reach it by asking the same questions?

Fig. 5 and Table IV show the top two levels of a depth-3 entropy tree fitted independently on each condition. Every internal node of all four trees is in Appendix B.

Four observations, in increasing order of interest.

The primary evidence is codec-invariant. All four trees select prosody_intensity_std at the root and extract almost

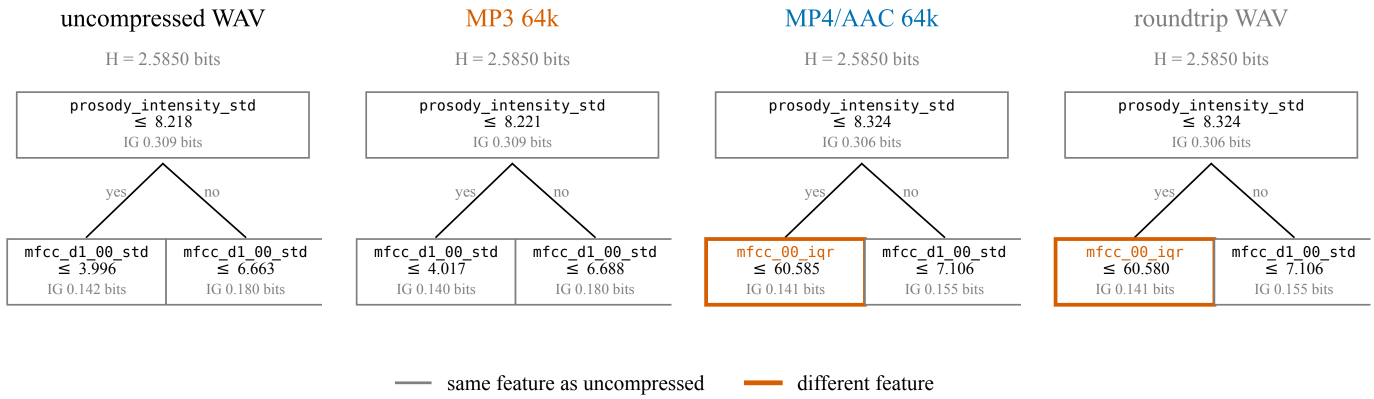

Fig. 5. The top two levels of the depth-3 entropy tree, fitted separately on each condition with the same actors and the same items. The root test is codec-invariant in identity and near-invariant in threshold and information gain. Divergence begins one level down, and it begins under AAC rather than under MP3: the AAC and roundtrip trees abandon the frame-aligned delta descriptor mfcc_d1_00_std for the static dispersion measure mfcc_00_iqr, which is what the 32.68 ms priming shift predicts. AAC and roundtrip — the same codec output through two containers — select identical descriptors with thresholds agreeing to four decimal places.

exactly the same information from it: 0.3090 bits under WAV and MP3, 0.3064 under AAC. Whatever coding does to this feature space, it does not change the single most discriminative question about a clip.

The threshold moves, and it moves with the mechanism. MP3 shifts the root threshold by 0.03 % (8.2184 $\rightarrow$ 8.2210); AAC shifts it by 1.29 % ($\rightarrow$ 8.3241). Intensity standard deviation is computed over frames, and AAC's priming adds 32.68 ms of near-silent lead-in to every clip, which raises the dispersion of the intensity contour. The tree compensates by raising its threshold. This is the padding mechanism of Fig. 2(c) surfacing as a number in a decision rule.

Divergence appears at depth 1, under AAC and not under MP3. The uncompressed and MP3 trees both ask about mfcc_d1_00_std, the standard deviation of the first temporal derivative of MFCC-0. The AAC and roundtrip trees abandon it for mfcc_00_iqr, a static dispersion measure of the same coefficient. That is precisely what a frame-alignment perturbation should do: it corrupts derivatives while leaving the distribution of the underlying contour intact, so the tree switches from a temporal statistic to a positional one at almost identical information gain (0.1408 vs 0.1415 bits). The model finds an equally good question that the codec has not damaged.

The container control reproduces at the structural level. mp4_aac64 and roundtrip_wav carry the same codec output through different containers. Their trees select identical descriptors at every node of the top two levels, with thresholds agreeing to 8.3241 vs 8.3241 at the root and 60.5853 vs 60.5802 at depth 1. Two conditions that differ only in container produce the same model; two conditions that differ in codec do not. The container/codec distinction, previously visible only as a difference in accuracy, is here visible as a difference in the learned decision rule.

E. Null calibration of the importance ranking

The obvious next step is to compare the models' full importance rankings across conditions, and the obvious result is dramatic: the tree's top-25 descriptors under MP3 share only 12 members with its top-25 under WAV, and the rank correlation over the descriptors either tree uses is $\rho = 0.017$ — no relationship at all. Read naively, coding has almost completely rewritten what the model considers important.

TABLE V
Null calibration for the decision tree's importance ranking. Rows above the rule are manipulations carrying no coding information; rows below are the coding conditions. ρ is Spearman rank correlation over the descriptors at least one of the two fits actually used.

Comparison	top-25 kept	ρ	What it varies
Tie-breaking	23.5	+0.946	estimator only; data fixed
Actor bootstrap	8.0	−0.200	training actors; codec fixed
Container re-wrap	16.0	+0.184	decode path; codec output fixed
MP3 @ 64k	12.0	+0.017	codec
MP4/AAC @ 64k	9.0	−0.082	codec
Roundtrip WAV	10.0	−0.123	codec

That reading is not supportable, and establishing why is this paper's methodological contribution. A decision tree's deep importance ranking is a fragile object. We measure three floors, all on the estimator that produced the effect.

Table V and Fig. 6(a) show why the naive reading fails. Resampling which actors the tree is fitted on — with the codec held fixed at uncompressed — retains only 8.0 of 25 descriptors, worse than any codec condition. Marginally compared, every codec effect sits inside the sampling floor, and no codec claim can be made at all.

The marginal comparison is the wrong one, because it varies two things at once: the floor is estimated on one training set and the effect on another. We therefore measure both inside the same draw. On each of 20 draws, 51 of the 60 training actors are sampled without replacement; the tree is fitted twice on those actors' uncompressed audio under different tie-breaking, giving that draw's floor, and once on the same actors' coded audio, giving that draw's effect. Sampling variance is thereby held fixed by

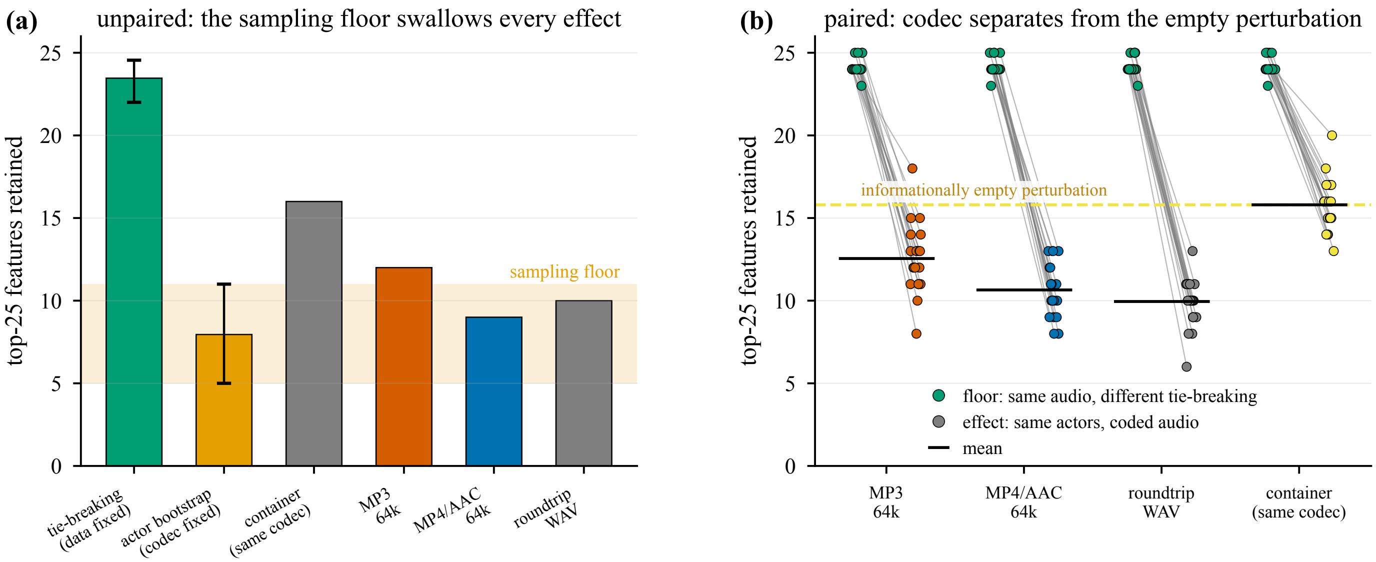

Fig. 6. A codec effect on a feature-importance ranking is uninterpretable without a null. (a) Compared marginally, the actor-resampling floor is so wide that every codec effect falls inside it. (b) Measured inside the same resampled training set — 20 draws of 51 of the 60 training actors, with the floor and the effect computed on the same draw — the codec effect separates cleanly. The decisive control is the rightmost group: an informationally empty container re-wrap already costs 8.4 of 25 descriptors, so only the gap between that line and the codec bars is codec-attributable.

TABLE VI
Paired within-draw comparison, 20 draws of 51 of 60 training actors. The floor is a refit on the same actors' uncompressed audio under different tie-breaking. Wilcoxon signed-rank on the paired per-draw difference.

	top-25 kept	ρ	p vs. floor
floor (same audio, refit)	24.15	+0.958	—
Container re-wrap (empty)	15.80	+0.196	8.1×10^{-5}
MP3 @ 64k	12.55	−0.068	8.4×10^{-5}
MP4/AAC @ 64k	10.65	−0.137	8.3×10^{-5}
Roundtrip WAV	9.95	−0.126	7.9×10^{-5}

construction, and no actor is duplicated.

Table VI and Fig. 6(b) give the result, and it has three layers.

The estimator is reproducible. With the data fixed and only tie-breaking varied, the tree retains 24.15 of 25 top descriptors ($\rho = 0.958$). Whatever else is fragile, the fitting procedure is not.

An informationally empty perturbation is expensive. The container re-wrap changes no codec information — it is the same AAC bitstream, decoded identically, differing only by one int16 requantisation whose median feature effect is 7.6×10^{-5} standardised units. It nonetheless costs 8.35 of 25 top descriptors ($24.15 \to 15.80$, $p = 8.1 \times 10^{-5}$). A ranking that turns over by a third under a manipulation that carries no information cannot be read as a description of what the model needs.

The codec effect is real but roughly two-fifths the size it appears. Against the empty-perturbation floor of 15.80, MP3 retains 12.55 ($p = 1.1 \times 10^{-3}$, $d_z = 1.20$) and AAC 10.65 ($p = 1.2 \times 10^{-4}$, $d_z = 2.39$). So of the 13.5 descriptors that AAC appears to displace relative to a clean refit, 8.35 are attributable to generic perturbation sensitivity and only 5.15 to the codec. A study reporting the raw figure would over-attribute by a factor of about 2.6.

The direction of the codec-specific component is consistent with Section VI-D: AAC displaces more than MP3 in the ranking (10.65 vs 12.55, $p = 4.2 \times 10^{-3}$, $d_z = 0.84$, paired), which is the reverse of the accuracy ordering under mismatch, where MP3 is catastrophic and AAC mild. Coding that perturbs every temporal derivative slightly reorganises a tree more than coding that blows out one band the tree can simply threshold around — which is also what Section VI-D shows structurally, since it is the AAC tree and not the MP3 tree that replaces its depth-1 descriptor.

F. Post-hoc attribution and its disagreements

Attribution was computed on the held-out speakers using TreeSHAP for the ensembles [11], LinearSHAP for the linear models, LIME for local surrogates [9], permutation importance, partial dependence, a depth-4 global surrogate tree, and, for the network, six methods from Captum [14]: Saliency, Input×Gradient, Integrated Gradients [12], DeepLIFT [13], GradientSHAP and Feature Ablation.

1) Convergent evidence across families: On uncompressed audio, five model families and four attribution algorithms select the same acoustic evidence. The top SHAP descriptors for XGBoost and LightGBM are identical in the first five (`mfcc_d1_00_std`, `mfcc_d2_00_std`, `duration_s`, `prosody_intensity_std`, `mfcc_00_std`); the decision tree leads with `prosody_intensity_std`, matching its root split; and all six gradient methods on the network return `mfcc_d1_00_std`, `mfcc_d2_00_std`, `prosody_cpps` and

TABLE VII
Share of total attribution mass by descriptor family, on uncompressed audio. Column share is the proportion of the 436 descriptors each family contributes. SHAP except the ANN, which uses integrated gradients.

Model	MFCC	spectral	prosody	chroma	global
column share	0.550	0.261	0.128	0.055	0.005
XGBoost	0.566	0.207	0.178	0.033	0.016
LightGBM	0.550	0.209	0.189	0.030	0.021
Random forest	0.438	0.315	0.205	0.032	0.011
Decision tree	0.436	0.183	0.306	0.034	0.041
Logistic regression	0.471	0.306	0.156	0.062	0.005
ANN (int. grad.)	0.542	0.238	0.153	0.057	0.009

TABLE VIII
Attribution stability under mismatch: Spearman ρ over all 436 descriptors against the uncompressed condition, with the number of top-25 descriptors retained in parentheses. SHAP for all rows except the ANN, which uses integrated gradients.

Model	mp3_64	mp4_aac64	roundtrip	Δ acc. MP3
Decision tree	0.995 (24)	0.988 (21)	0.988 (21)	−0.009
XGBoost	0.991 (25)	0.984 (22)	0.984 (22)	+0.004
Random forest	0.990 (25)	0.976 (25)	0.975 (25)	−0.003
LightGBM	0.988 (23)	0.983 (22)	0.982 (22)	−0.005
Logistic regression	0.987 (22)	0.942 (21)	0.941 (21)	**−0.334**
ANN (int. grad.)	0.677 (11)	0.891 (16)	0.887 (16)	−0.246

duration_s. The two leaders are the standard deviations of the first and second derivative of MFCC-0 — how variable the rate of change of overall loudness is — a direct formalisation of vocal agitation.

Attribution mass is spread across families rather than concentrated (Table VII), and the pattern matches the mutual-information result of Section IV: MFCCs hold 55 % of the columns and take roughly 55 % of the attribution, so they carry no per-descriptor advantage. The single decision tree is the outlier, leaning harder on prosody — which is what makes it readable and also what caps its accuracy.

Per-emotion attributions are acoustically sensible. CPPS, a breathiness correlate, drives both sadness and neutrality; F_0 median drives fear; F_2 movement drives disgust; and loudness dynamics drive anger. The network was told none of this.

2) The methods disagree with each other: Fig. 7(a) is the uncomfortable result. SHAP agrees closely with a model's intrinsic importance for the simpler models ($\rho = 0.98$ for both the decision tree and logistic regression), but LIME and permutation importance are close to unrelated to it. On the decision tree, LIME and SHAP rank descriptors at $\rho = 0.008$ — no relationship at all — while still sharing 10 of their top 20. This reproduces the disagreement problem Krishna et al. [7] document, on a model simple enough that no approximation error can be blamed.

Panel (b) sharpens the point rather than softening it. The six gradient-family methods on the network agree almost perfectly ($\rho \geq 0.94$ pairwise, with Integrated Gradients and DeepLIFT at 0.998). High agreement within a method family is therefore not evidence of correctness — these methods share a mathematical lineage, and their consensus reflects that lineage. The practical consequence is that a single-method claim about which acoustic descriptors matter is not reproducible across methods. Only descriptors surviving all four independent views should be reported as robust; here that set is mfcc_d1_00_std, prosody_intensity_std, duration_s, prosody_cpps and prosody_f0_median.

Two further calibrations belong with this result. A depth-4 global surrogate reproduces its target model's predictions only 57 to 74 % of the time (random forest 73.9 %, LightGBM 63.4 %, XGBoost 62.1 %, SVM 58.7 %, logistic regression 57.1 %), so surrogate rules summarise a majority of a model, not a model; reporting them without their fidelity would overstate what has been explained. And the network's attributions pass the Adebayo model-randomisation check (Fig. 8): correlation between attributions from the trained network and a randomly initialised one of identical architecture ranges from −0.011 to 0.206 across all six methods. These explanations describe the trained model, not merely the input distribution.

3) An estimator that failed loudly: The RBF-SVM has no exact explainer, so it was given KernelSHAP with 100 sampled coalitions over 436 descriptors. KernelSHAP fits a weighted least-squares regression with one unknown per descriptor; with 100 coalitions for 436 unknowns that system is rank-deficient, and the result diverged — maximum mean |SHAP| of 2.1×10^{12} against a median of 5.3×10^{-4}, with entire descriptor families assigned exactly zero. A cheap tell that this had gone wrong: a well-behaved additive attribution over 7,441 clips does not assign a whole family exactly 0.000.

The SVM's KernelSHAP ranking is excluded from every conclusion in this paper as an estimation artefact, and the implementation now raises when the coalition budget does not exceed the descriptor count. The SVM contributes accuracy and robustness results but no attribution row; its trustworthy views are permutation importance and LIME, whose top descriptors reproduce the consensus every other model reaches.

G. Attribution stability under coding

Table VIII reports SHAP rank stability under train/serve mismatch. Tree and linear attributions barely move; the network's move substantially.

These rows measure something different from Section VI-E, and the distinction matters. Here the model is fixed and only its input changes, so the comparison is between two attributions of one fitted object. In Section VI-E the model is refitted, so the comparison is between two learned objects. The first is stable because a fitted tree's split structure does not move when you feed it slightly different numbers; the second is not, because the fitting procedure can reach an equally good structure by a different route.

Even so, the accuracy figure understates the attribution figure throughout. XGBoost loses 2.2 accuracy points on

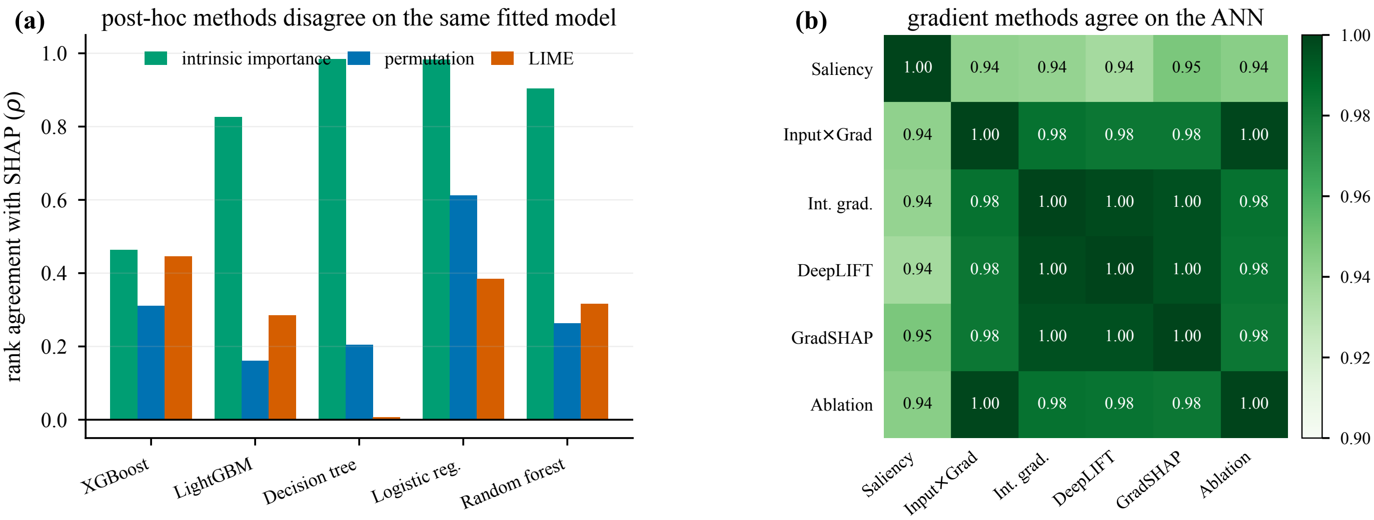

Fig. 7. Explanation-method agreement, measured as Spearman rank correlation over all 436 descriptors. (a) Post-hoc methods applied to the same fitted tabular model. SHAP tracks intrinsic importance closely for the simpler models but LIME and permutation importance are nearly unrelated to it; on the decision tree, LIME and SHAP agree at $\rho = 0.008$. (b) The six gradient-family methods applied to the network agree almost perfectly ($\rho \geq 0.94$ pairwise), which shows that the disagreement in (a) is a property of the method families rather than of this feature space.

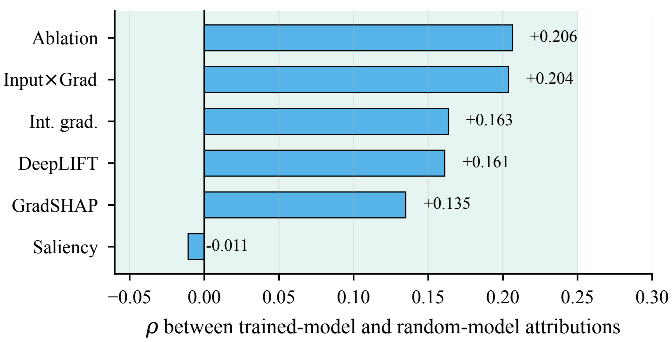

Fig. 8. Model-randomisation sanity check [6] for the six gradient-family methods. A method whose attributions correlate with those of a randomly initialised network of identical architecture is describing the input distribution rather than the model. All six sit near zero, so the failure mode does not apply here.

AAC and still swaps 3 of its top 25; the network loses 8.7 points and swaps 9. An accuracy-only benchmark would call the first negligible and the second modest.

What replaces what is readable only because the descriptors are named. Under MP3 the descriptors entering the network's top 25 are almost all encoder artefacts — spectral_flatness in five statistics, plus spectral_contrast_6_mean — while those leaving are genuine voice-quality and prosodic measures: prosody_cpps, prosody_f0_skew, prosody_f2_mean, prosody_f2_std, prosody_f3_bw_median. Under mismatch, the model stops listening to the voice and starts listening to the encoder.

1) A blind spot in global attribution metrics: The logistic regression row of Table VIII is the most important negative result in this paper. Its SHAP ranking is essentially unchanged under MP3 ($\rho = 0.987$, 22 of 25 top descriptors retained) while its balanced accuracy collapses

TABLE IX
Balanced accuracy on MP3 as the most codec-shifted descriptors are replaced by training medians. Italic row: the model's own clean-audio score. Tree columns are flat because they had nothing to recover.

Masked	SVM	Log. reg.	ANN	MLP	XGBoost	Dec. tree
clean	*0.586*	*0.567*	*0.599*	*0.574*	*0.582*	*0.391*
0	0.203	0.233	0.354	0.383	0.586	0.382
1	0.578	0.502	0.580	0.552	0.581	0.382
2	0.583	0.529	0.587	0.554	0.581	0.391
3	0.579	0.562	0.595	0.565	0.580	0.391
10	0.575	0.539	0.589	0.562	0.582	0.394
80	0.551	0.435	0.574	0.523	0.564	0.396

from 0.567 to 0.233. The coefficients did not change; only the inputs went out of range, and a global mean-|SHAP| ranking is largely blind to that.

This is a caution about the method, not about the model. A global attribution-stability metric can report "nothing changed" while the classifier has effectively stopped working. Any deployment that monitors explanation stability as a proxy for model health must pair it with input-range monitoring or local, per-instance attribution; the global ranking alone will not raise the alarm.

H. Diagnosis and repair

Because the descriptors are named, the mismatch failure is not merely observable but addressable. Descriptors were ranked by how far each codec moves them in units of training standard deviation, then progressively replaced with training medians in both the compressed and the clean test set — the clean column controlling for whatever real signal the masking itself destroys.

The single masked descriptor is spectral_contrast_6_mean. Masking it alone recovers 98 % of the RBF-SVM's 38.3-point loss and 92 % of the network's 24.6-point loss; three

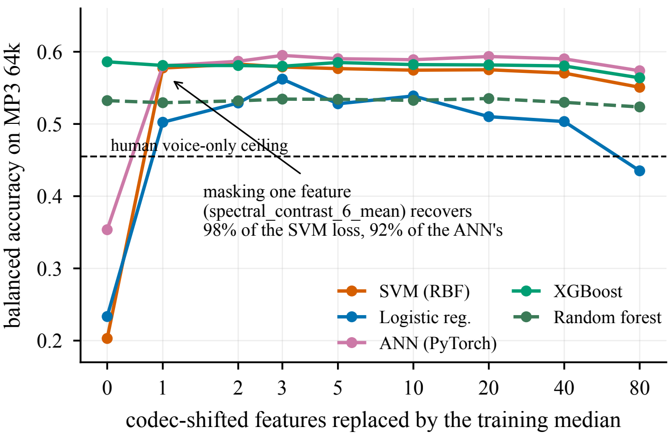

Fig. 9. Neutralisation on MP3 at 64 kbit s^{-1}: replacing the most codec-shifted descriptors with their training medians. Masking one descriptor recovers almost the entire collapse for every affected model; the tree models were never affected and are flat throughout. Beyond roughly 20 descriptors the masking begins to destroy real evidence. On uncompressed audio the same masking costs nothing out to 20 descriptors, confirming these columns were never load-bearing evidence.

descriptors bring logistic regression to 0.562 against its 0.567 clean-audio score.

Two controls make this a diagnosis rather than a coincidence. On uncompressed audio the identical masking costs nothing out to 20 descriptors — the network's clean control sits at 0.607–0.609, marginally above its 0.599 unmasked baseline — so these columns were never load-bearing evidence; they were the channel through which the encoder's fingerprint entered the model. And the AAC case behaves as its different mechanism predicts: no single descriptor dominates, with the network's best recovery arriving at 10 descriptors masked (0.513 → 0.560) and the SVM's at 40 (0.499 → 0.542).

The diagnosis is therefore covariate shift on a small set of identifiable descriptors, not loss of emotional information — which is exactly what Section VI-C establishes independently, by a different route.

VII. Discussion

A. Three questions that are usually one

This study separates three things that a single accuracy number conflates. Does the codec destroy the information? No: matched-format training recovers 38.6 points and lands within 0.003 of the clean score. Does the codec break deployed models? Severely, if they consume descriptor magnitudes and were fitted on clean audio. Does the codec change what the model learns? Yes, but by much less than the raw numbers suggest, and only a properly nulled comparison can say by how much.

An evaluation reporting only the second question — which is what cross-format benchmarking does — would conclude that 64 kbit s^{-1} MP3 is unusable for SER. The first question shows that conclusion is about the pipeline, not the codec.

B. Null calibration is not optional

The container re-wrap changes no coding information and still turns over a third of the tree's top-25 ranking. Any claim of the form "manipulation M changed which features the model uses" is therefore uninterpretable without a perturbation matched in every respect except informational content. We propose this as standard practice for feature-attribution claims in affective computing. The cost is one additional condition and one additional model fit; the benefit is that the claim becomes falsifiable, and in our case that it shrinks by a factor of 2.6 rather than being reported at face value.

The paired design matters as much as the null itself. Compared marginally, the actor-resampling floor was wide enough to swallow every codec effect (Fig. 6(a)) and we would have reported nothing. Measuring floor and effect inside the same resampled training set removes sampling variance by construction and recovers a clean answer.

C. What a threshold buys, and what it does not

Under mismatch, axis-aligned models are almost perfectly robust to a 38σ excursion, because a threshold test is invariant to how far past the threshold a value lands. That is a real and cheap engineering advantage. But Section VI-C shows it disappears under matched training, and Section VI-E shows the same models' learned structure is more sensitive to a small diffuse perturbation than a large concentrated one. The threshold protects the decision, not the evidence.

D. Auditing and the archive problem

Per-instance rationales are increasingly surfaced to operators and auditors. Our results indicate that such rationales are not invariant across a transcoding step that may leave the decision invariant. Where an audit is conducted on archived compressed material while inference runs on uncompressed input — or the converse — the decisions will agree while the explanations will not. An explanation is not reproducible unless the delivery format is part of the record, and to our knowledge no SER benchmark currently reports it.

E. What named descriptors buy

The repair in Section VI-H has no analogue in a system whose front end is learned. Identifying spectral_contrast_6_mean as the contaminated channel required attribution over units with acoustic meaning; neutralising it required reaching into the representation. So did reading the codec's frame-alignment mechanism off a decision threshold (Section VI-D).

This is a narrower claim than Rudin's [16]. We are not arguing that interpretable models should replace strong ones — Table I shows the single interpretable model in our zoo costing a third of achievable performance, and the best model overall is a neural network. We are arguing that a named-descriptor front end retains a diagnostic and repair capability that end-to-end pipelines lack, and that in codec-heterogeneous deployments this capability has measurable value: 98 % of a 38-point loss, recovered by masking one column.

VIII. Limitations

1) Acted, not spontaneous, emotion. CREMA-D actors perform prescribed emotions on 12 fixed sentences. Absolute accuracies do not transfer to spontaneous speech, and the human voice-only ceiling of 45.5 % shows the labels are themselves only loosely recoverable from audio.
2) One split, not repeated cross-validation, for the headline tables. Tables I–III come from a single deterministic 60/11/20 actor partition. Differences of a point or two should not be over-read. The null-calibration analysis of Section VI-E does resample (20 draws), and is the only place we make a claim that depends on resampling.
3) Two codecs, one bitrate. Only $64\,\mathrm{kbit\,s^{-1}}$ MP3 and AAC were tested. The MP3 mismatch result is severe partly because $64\,\mathrm{kbit\,s^{-1}}$ is aggressive for that codec; higher bitrates would likely show a gradient, and Opus — now the dominant conversational codec — is untested.
4) Summary statistics discard time. Every frame contour is collapsed to eight order statistics, so no attribution here localises when in a clip the evidence sits. The AAC padding mechanism is inferred from duration and from which descriptors move, not observed directly in time.
5) The container control is not a perfect null. roundtrip_wav differs from mp4_aac64 by an int16 requantisation, so it is an almost empty perturbation rather than an exactly empty one. It is conservative in the right direction — a truly empty perturbation would have a lower floor and make the codec effects look larger — but a bit-exact container null would be a stronger instrument.
6) The null calibration covers one estimator. Section VI-E calibrates the decision tree, chosen because it is the paper's readable model and the one whose ranking is most obviously fragile. Whether ensemble or gradient attributions need floors of the same size is not established here, though Table VIII's fixed-model comparison suggests they are more stable.
7) Attribution comparisons are global. Spearman ρ and top-k overlap over mean-|attribution| are coarse instruments, and Section VI-G1 shows they can miss a total accuracy collapse. A local, per-instance stability analysis would be stronger and is not attempted.
8) Speaker-dominance limits what importance rankings can mean. With mean $\eta^2 = 0.178$ for speaker against 0.098 for emotion, a substantial part of any importance ranking describes which voices are in the training set. The actor-bootstrap floor in Table V quantifies this and it is large.
9) Exploratory, not confirmatory. The per-condition and null-calibration analyses were designed after inspecting the cross-format results. Confirmatory status requires preregistered replication.

IX. Future Work

Three directions follow from specific results. A bit-exact container null — achievable by comparing two lossless containers carrying identical PCM — would tighten the floor of Section VI-E and is cheap. Extending the null calibration to SHAP and integrated-gradients rankings under refitting would establish whether the fragility we document is specific to trees or general to feature attribution. And a bitrate ladder, with Opus alongside MP3 and AAC, would turn the two-point comparison in Section VI-D into a dose-response curve for structural displacement.

Beyond emotion, speaker identity, vocal stress and voice quality are carried by precisely the spectral detail perceptual coding discards, and the matched-format result may not transfer: a task whose evidence lives in the top octave has less to fall back on than one whose evidence lives in loudness dynamics.

X. Conclusion

We asked whether perceptual audio coding changes what a speech emotion model relies on, using models whose evidence is a named acoustic descriptor rather than a latent dimension.

Coding does not remove the affective information: a kernel machine that scores 0.203 on MP3 when fitted to clean audio scores 0.589 when fitted to MP3 itself, against 0.586 on clean audio. The catastrophic cross-format losses in the literature and in our own Table II are train/serve mismatch, and they are repairable either by matched training or by neutralising a single contaminated descriptor.

What coding does change is the structure the model learns. An entropy tree keeps the same root question in every condition and extracts the same information from it, but AAC moves the root threshold by 1.3 % and replaces the depth-1 descriptor with one its frame-alignment perturbation leaves intact — while the same two conditions carrying identical codec output through different containers produce identical trees.

The size of that change is easy to overstate, and the discipline that prevents it is the paper's most transportable result. An informationally empty container rewrap already displaces 8.35 of a tree's 25 most important descriptors. Only the gap beyond that floor — 3.25 descriptors for MP3, 5.15 for AAC — is codec-attributable. We would ask reviewers of affective computing systems to require, alongside any claim that an intervention changed a model's explanation, the divergence induced by an intervention carrying no information. Without it, the claim cannot be evaluated; with it, ours shrinks by a factor of two and a half and becomes defensible.

Data Availability Statement

CREMA-D is publicly available from its authors under the terms stated in [22]. All derived artefacts are released with this paper: the stimulus construction scripts and condition manifests, the JSON schema of the 436-descriptor table over 29,768 rendered files, the per-condition and null-calibration result files, and the JSON/CSV files from which every table and figure is generated. The figure- and appendix-generation scripts read only these committed artefacts and hard-code no numbers. The feature table itself (243 MB) and the rendered audio are regenerable from the committed extraction code and configuration.

Ethics Declaration

This study involves no human subjects and no personally identifying data. It uses a published corpus of consenting professional actors performing scripted sentences, collected and released for research use by its original authors under their own ethical approval [22]. No new recordings were made. We note that speech emotion recognition is a dual-use technology; the findings here concern robustness and explanation validity and are, if anything, cautionary about deploying such systems on heterogeneous audio without format-aware validation. Nothing here should be read as evidence that automatic emotion inference is accurate, fair, or appropriate in consequential settings.

Author Contributions (CRediT)

Akshath R: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing — original draft, Writing — review & editing, Visualization, Project administration.

Conflict of Interest Statement

The author declares no competing financial or non-financial interests.

Funding

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors. Compute costs were borne by the author.

Declaration of Generative AI Use

Generative AI tools were used to assist with code implementation, figure generation, and manuscript drafting. All experimental design decisions, all analyses, and all interpretations are the author's. Every numerical value reported was produced by the committed pipeline and traced to its source artefact. The author takes full responsibility for the content.

Appendix A
Complete Feature Inventory

Every stimulus is reduced to 436 named columns. Frame-level contours are summarised by eight order statistics (mean, standard deviation, minimum, maximum, median, interquartile range, skewness, excess kurtosis); descriptors marked with a shorter list are summarised only by those statistics, and scalars are already clip-level. Naming is <family>_<descriptor>_<statistic> throughout, which is what makes an attribution over a column a claim about an acoustic quantity.

Analysis frames are 512 point FFT, 400 sample window, 160 sample hop at 16 kHz — 25 ms windows every 10 ms. Praat measures use a 75 Hz to 500 Hz pitch range.

A. Descriptors and their definitions

B. Enumerated column names

The complete inventory, in the order the extractor emits it.

Appendix B
Per-Condition Tree Splits

Every internal node of the depth-3 entropy tree of Section VI-D, fitted independently on each condition with the same actors, the same items and the same hyperparameters. Path reads from the root: `L` is the branch taken when the parent's test is satisfied. Entropies are in bits; n is the raw sample count and the information gain is computed on the class-weighted counts (Eq. 3).

References

[1] A. P. Reddy and V. Vijayarajan, "Audio compression with multi-algorithm fusion and its impact in speech emotion recognition," International Journal of Speech Technology, vol. 23, no. 2, pp. 277–285, 2020.
[2] A. Ghorbani, A. Abid, and J. Zou, "Interpretation of neural networks is fragile," arXiv:1710.10547, 2019.
[3] D. Alvarez-Melis and T. S. Jaakkola, "On the robustness of interpretability methods," arXiv:1806.08049, 2018.
[4] F. Eyben, K. R. Scherer, B. W. Schuller, J. Sundberg, E. André, C. Busso, L. Y. Devillers, J. Epps, P. Laukka, S. S. Narayanan, and K. P. Truong, "The Geneva minimalistic acoustic parameter set (GeMAPS) for voice research and affective computing," IEEE Transactions on Affective Computing, vol. 7, no. 2, pp. 190–202, 2016.
[5] F. Eyben, M. Wöllmer, and B. Schuller, "openSMILE: The Munich versatile and fast open-source audio feature extractor," in Proc. 18th ACM Int. Conf. Multimedia, 2010, pp. 1459–1462.
[6] J. Adebayo, J. Gilmer, M. Muelly, I. Goodfellow, M. Hardt, and B. Kim, "Sanity checks for saliency maps," arXiv:1810.03292, 2018.
[7] S. Krishna, T. Han, A. Gu, S. Wu, S. Jabbari, and H. Lakkaraju, "The disagreement problem in explainable machine learning: A practitioner's perspective," arXiv:2202.01602, 2022.
[8] H. Wu, X. Chen, Y.-C. Lin, K. Chang, J. Du, K.-H. Lu, A. H. Liu, H.-L. Chung, Y.-K. Wu, D. Yang, S. Liu, Y.-C. Wu, X. Tan, J. Glass, S. Watanabe, and H.-y. Lee, "Codec-SUPERB @ SLT 2024: A lightweight benchmark for neural audio codec models," arXiv:2409.14085, 2024.
[9] M. T. Ribeiro, S. Singh, and C. Guestrin, ""Why should I trust you?": Explaining the predictions of any classifier," in Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining, 2016, pp. 1135–1144.
[10] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," arXiv:1705.07874, 2017.

TABLE X
Spectral descriptors — 114 columns of 436.

Descriptor	Definition	Statistics	n
Spectral centroid	First moment of the power spectrum; the frequency about which spectral energy is balanced. Correlates with perceived brightness.	mean, std, min, max, median, IQR, skew, kurtosis	8
Spectral bandwidth	Power-weighted second moment about the centroid; how widely energy is spread in frequency.	mean, std, min, max, median, IQR, skew, kurtosis	8
Spectral roll-off (85 %)	Frequency below which 85 % of the frame's spectral energy lies.	mean, std, min, max, median, IQR, skew, kurtosis	8
Spectral roll-off (95 %)	As above at 95 %. Directly sensitive to codec low-pass behaviour.	mean, std, min, max, median, IQR, skew, kurtosis	8
Spectral flatness	Geometric mean of the magnitude spectrum divided by its arithmetic mean. Approaches 1 for noise-like frames and 0 for tonal ones; collapses when an encoder zeroes masked bins.	mean, std, min, max, median, IQR, skew, kurtosis	8
Zero-crossing rate	Fraction of adjacent sample pairs with a sign change; a cheap voicing and frication proxy.	mean, std, min, max, median, IQR, skew, kurtosis	8
Frame RMS energy	Root-mean-square magnitude per frame; the loudness contour.	mean, std, min, max, median, IQR, skew, kurtosis	8
Spectral flux	Euclidean norm of the frame-to-frame difference of the magnitude spectrum; how fast the spectrum is changing.	mean, std, min, max, median, IQR, skew, kurtosis	8
Spectral entropy	Shannon entropy of the power spectrum normalised to its own bin count (Eq. 1); a scale-free measure of how evenly energy is distributed across frequency.	mean, std, min, max, median, IQR, skew, kurtosis	8
Spectral slope	Least-squares slope of magnitude regressed on frequency, per frame; spectral tilt.	mean, std, min, max, median, IQR, skew, kurtosis	8
Band-energy ratios (8 bands)	Fraction of frame energy in each of 0–500, 500–1000, 1000–2000, 2000–3000, 3000–4000, 4000–5000, 5000–6000, 6000–8000 Hz. Energy-normalised, so these describe spectral shape rather than level.	mean, std	16
High-frequency survival ratios	Fraction of frame energy above 4 and above 6 kHz. The most direct probes of lossy-codec low-pass behaviour in the inventory.	mean, std	4
Spectral contrast (7 sub-bands)	Per-band difference between spectral peaks and valleys in decibels. Band 6 covers 6–8 kHz and is the single most codec-displaced column in the table (Section III-C).	mean, std	14

TABLE XI
Cepstral descriptors — 240 columns of 436.

Descriptor	Definition	Statistics	n
ΔMFCC (20)	First-order temporal derivative of each MFCC, over a 9-frame window.	mean, std	40
ΔΔMFCC (20)	Second-order temporal derivative of each MFCC.	mean, std	40
MFCC (20)	Discrete cosine transform of the log mel-filterbank energies; the standard SER front end. MFCC-0 tracks overall log energy, so its derivatives measure how fast loudness changes.	mean, std, min, max, median, IQR, skew, kurtosis	160

TABLE XII
Prosodic and voice-quality descriptors (Praat) — 56 columns of 436.

Descriptor	Definition	Statistics	n
F_0 slope (mean $\lvert\cdot\rvert$)	Mean absolute first difference of the voiced F_0 contour, per second; pitch movement rate.	scalar	1
Fundamental frequency F_0	Praat autocorrelation pitch over voiced frames only.	mean, std, min, max, median, IQR, skew, kurtosis	8
Jitter (4 variants)	Cycle-to-cycle variation in glottal period: local, RAP, PPQ5, DDP. A perturbation measure of the voice source.	scalar	4
Shimmer (6 variants)	Cycle-to-cycle variation in glottal amplitude: local, local dB, APQ3, APQ5, APQ11, DDA. APQ11 needs eleven consecutive periods and is the one column in the table with non-trivial missingness (11.3 %).	scalar	6
Harmonics-to-noise ratio	Ratio of periodic to aperiodic energy in decibels; a hoarseness and breathiness correlate.	mean, std, min, max, median	5
Formant bandwidths B_1–B_3	Median −3 dB bandwidth of the first three formants (Burg estimator).	median	1
Formants F_1–F_5	Burg-estimator formant frequencies; the vocal-tract filter. F_2 movement is the strongest single correlate of disgust in this corpus.	mean, std, median	17
Intensity	Praat short-term intensity contour in decibels. Its standard deviation is among the top-ranked features for every model family in the study.	mean, std, min, max, median, IQR, skew, kurtosis	8
Voiced fraction	Proportion of frames with a defined F_0.	scalar	1
Pause fraction	Complement of the voiced fraction; definitionally 1− voiced fraction ($r = -1.00$).	scalar	1
Voiced-segment count	Number of maximal runs of voiced frames.	scalar	1
Voiced-segment rate	Voiced segments per second; a speech-rate proxy.	scalar	1
Mean voiced-segment duration	Mean length in seconds of a voiced run.	scalar	1
CPPS	Smoothed cepstral peak prominence; the standard acoustic correlate of breathy phonation. Drives both sadness and neutrality in every attribution method applied here.	scalar	1

[11] S. M. Lundberg, G. Erion, H. Chen, A. DeGrave, J. M. Prutkin, B. Nair, R. Katz, J. Himmelfarb, N. Bansal, and S.-I. Lee, "From local explanations to global understanding with explainable AI for trees," Nature Machine Intelligence, vol. 2, no. 1, pp. 56–67, 2020.

TABLE XIII
Chroma descriptors — 24 columns of 436.

Descriptor	Definition	Statistics	n
Chroma (12 pitch classes)	Energy folded onto the twelve semitone pitch classes. Tuning is pinned to 0 rather than estimated, because an estimated tuning could itself shift under compression and become part of the effect being measured.	mean, std	24

TABLE XIV
Global descriptors — 2 columns of 436.

Descriptor	Definition	Statistics	n
Clip duration	Decoded duration in seconds. Rises by 32.68 ms under AAC encoder priming, uniformly across every clip.	scalar	1
Peak amplitude	Maximum absolute sample value after loudness normalisation.	scalar	1

[12] M. Sundararajan, A. Taly, and Q. Yan, "Axiomatic attribution for deep networks," arXiv:1703.01365, 2017.

[13] A. Shrikumar, P. Greenside, and A. Kundaje, "Learning important features through propagating activation differences," arXiv:1704.02685, 2017.

[14] N. Kokhlikyan, V. Miglani, M. Martin, E. Wang, B. Alsallakh, J. Reynolds, A. Melnikov, N. Kliushkina, C. Araya, S. Yan, and O. Reblitz-Richardson, "Captum: A unified and generic model interpretability library for PyTorch," arXiv:2009.07896, 2020.

[15] R. Guidotti, A. Monreale, S. Ruggieri, F. Turini, F. Giannotti, and D. Pedreschi, "A survey of methods for explaining black box models," ACM Computing Surveys, vol. 51, no. 5, pp. 1–42, 2018.

[16] C. Rudin, "Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead," Nature Machine Intelligence, vol. 1, no. 5, pp. 206–215, 2019.

[17] V. Haunschmid, E. Manilow, and G. Widmer, "audioLIME: Listenable explanations using source separation," arXiv:2008.00582, 2020.

[18] S. Becker, J. Vielhaben, M. Ackermann, K.-R. Müller, S. Lapuschkin, and W. Samek, "AudioMNIST: Exploring explainable artificial intelligence for audio analysis on a simple benchmark," arXiv:1807.03418, 2018.

[19] T. Sotirou, V. Lyberatos, O. Menis Mastromichalakis, and G. Stamou, "MusicLIME: Explainable multimodal music understanding," arXiv:2409.10496, 2024.

[20] S. Nasr, Z. Ren, and D. Johnson, "Beyond saliency: Enhancing explanation of speech emotion recognition with expert-referenced acoustic cues," arXiv:2511.11691, 2025.

[21] J. G. Moreno-Torres, T. Raeder, R. Alaiz-Rodríguez, N. V. Chawla, and F. Herrera, "A unifying view on dataset shift in classification," Pattern Recognition, vol. 45, no. 1, pp. 521–530, 2012.

[22] H. Cao, D. G. Cooper, M. K. Keutmann, R. C. Gur, A. Nenkova, and R. Verma, "CREMA-D: Crowd-sourced emotional multimodal actors dataset," IEEE Transactions on Affective Computing, vol. 5, no. 4, pp. 377–390, 2014.

[23] Y. Jadoul, B. Thompson, and B. de Boer, "Introducing Parselmouth: A Python interface to Praat," Journal of Phonetics, vol. 71, pp. 1–15, 2018.

[24] B. McFee, C. Raffel, D. Liang, D. P. W. Ellis, M. McVicar, E. Battenberg, and O. Nieto, "librosa: Audio and music signal analysis in Python," in Proc. 14th Python in Science Conference, 2015, pp. 18–24.

[25] L. Breiman, "Random forests," Machine Learning, vol. 45, no. 1, pp. 5–32, 2001.

[26] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining, 2016, pp. 785–794.

[27] C. Cortes and V. Vapnik, "Support-vector networks," Machine Learning, vol. 20, no. 3, pp. 273–297, 1995.

[28] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and É. Duchesnay, "Scikit-learn: Machine learning in Python," arXiv:1201.0490, 2011, journal of Machine Learning Research, vol. 12, pp. 2825–2830.

TABLE XV
Spectral descriptors: all 114 column names.

spectral__centroid__mean	spectral__rolloff95__iqr	spectral__flux__min	spectral__band__2000__3000__std
spectral__centroid__std	spectral__rolloff95__skew	spectral__flux__max	spectral__band__3000__4000__mean
spectral__centroid__min	spectral__rolloff95__kurt	spectral__flux__median	spectral__band__3000__4000__std
spectral__centroid__max	spectral__flatness__mean	spectral__flux__iqr	spectral__band__4000__5000__mean
spectral__centroid__median	spectral__flatness__std	spectral__flux__skew	spectral__band__4000__5000__std
spectral__centroid__iqr	spectral__flatness__min	spectral__flux__kurt	spectral__band__5000__6000__mean
spectral__centroid__skew	spectral__flatness__max	spectral__entropy__mean	spectral__band__5000__6000__std
spectral__centroid__kurt	spectral__flatness__median	spectral__entropy__std	spectral__band__6000__8000__mean
spectral__bandwidth__mean	spectral__flatness__iqr	spectral__entropy__min	spectral__band__6000__8000__std
spectral__bandwidth__std	spectral__flatness__skew	spectral__entropy__max	spectral__hf_ratio__4000__mean
spectral__bandwidth__min	spectral__flatness__kurt	spectral__entropy__median	spectral__hf_ratio__4000__std
spectral__bandwidth__max	spectral__zcr__mean	spectral__entropy__iqr	spectral__hf_ratio__6000__mean
spectral__bandwidth__median	spectral__zcr__std	spectral__entropy__skew	spectral__hf_ratio__6000__std
spectral__bandwidth__iqr	spectral__zcr__min	spectral__entropy__kurt	spectral__contrast__0__mean
spectral__bandwidth__skew	spectral__zcr__max	spectral__slope__mean	spectral__contrast__0__std
spectral__bandwidth__kurt	spectral__zcr__median	spectral__slope__std	spectral__contrast__1__mean
spectral__rolloff85__mean	spectral__zcr__iqr	spectral__slope__min	spectral__contrast__1__std
spectral__rolloff85__std	spectral__zcr__skew	spectral__slope__max	spectral__contrast__2__mean
spectral__rolloff85__min	spectral__zcr__kurt	spectral__slope__median	spectral__contrast__2__std
spectral__rolloff85__max	spectral__rms__mean	spectral__slope__iqr	spectral__contrast__3__mean
spectral__rolloff85__median	spectral__rms__std	spectral__slope__skew	spectral__contrast__3__std
spectral__rolloff85__iqr	spectral__rms__min	spectral__slope__kurt	spectral__contrast__4__mean
spectral__rolloff85__skew	spectral__rms__max	spectral__band__0__500__mean	spectral__contrast__4__std
spectral__rolloff85__kurt	spectral__rms__median	spectral__band__0__500__std	spectral__contrast__5__mean
spectral__rolloff95__mean	spectral__rms__iqr	spectral__band__500__1000__mean	spectral__contrast__5__std
spectral__rolloff95__std	spectral__rms__skew	spectral__band__500__1000__std	spectral__contrast__6__mean
spectral__rolloff95__min	spectral__rms__kurt	spectral__band__1000__2000__mean	spectral__contrast__6__std
spectral__rolloff95__max	spectral__flux__mean	spectral__band__1000__2000__std	
spectral__rolloff95__median	spectral__flux__std	spectral__band__2000__3000__mean	

TABLE XVI
Cepstral descriptors: all 240 column names.

mfcc__00__mean	mfcc__07__median	mfcc__15__mean	mfcc__d1__05__mean
mfcc__00__std	mfcc__07__iqr	mfcc__15__std	mfcc__d1__05__std
mfcc__00__min	mfcc__07__skew	mfcc__15__min	mfcc__d2__05__mean
mfcc__00__max	mfcc__07__kurt	mfcc__15__max	mfcc__d2__05__std
mfcc__00__median	mfcc__08__mean	mfcc__15__median	mfcc__d1__06__mean
mfcc__00__iqr	mfcc__08__std	mfcc__15__iqr	mfcc__d1__06__std
mfcc__00__skew	mfcc__08__min	mfcc__15__skew	mfcc__d2__06__mean
mfcc__00__kurt	mfcc__08__max	mfcc__15__kurt	mfcc__d2__06__std
mfcc__01__mean	mfcc__08__median	mfcc__16__mean	mfcc__d1__07__mean
mfcc__01__std	mfcc__08__iqr	mfcc__16__std	mfcc__d1__07__std
mfcc__01__min	mfcc__08__skew	mfcc__16__min	mfcc__d2__07__mean
mfcc__01__max	mfcc__08__kurt	mfcc__16__max	mfcc__d2__07__std
mfcc__01__median	mfcc__09__mean	mfcc__16__median	mfcc__d1__08__mean
mfcc__01__iqr	mfcc__09__std	mfcc__16__iqr	mfcc__d1__08__std
mfcc__01__skew	mfcc__09__min	mfcc__16__skew	mfcc__d2__08__mean
mfcc__01__kurt	mfcc__09__max	mfcc__16__kurt	mfcc__d2__08__std
mfcc__02__mean	mfcc__09__median	mfcc__17__mean	mfcc__d1__09__mean
mfcc__02__std	mfcc__09__iqr	mfcc__17__std	mfcc__d1__09__std
mfcc__02__min	mfcc__09__skew	mfcc__17__min	mfcc__d2__09__mean
mfcc__02__max	mfcc__09__kurt	mfcc__17__max	mfcc__d2__09__std
mfcc__02__median	mfcc__10__mean	mfcc__17__median	mfcc__d1__10__mean
mfcc__02__iqr	mfcc__10__std	mfcc__17__iqr	mfcc__d1__10__std
mfcc__02__skew	mfcc__10__min	mfcc__17__skew	mfcc__d2__10__mean
mfcc__02__kurt	mfcc__10__max	mfcc__17__kurt	mfcc__d2__10__std
mfcc__03__mean	mfcc__10__median	mfcc__18__mean	mfcc__d1__11__mean
mfcc__03__std	mfcc__10__iqr	mfcc__18__std	mfcc__d1__11__std
mfcc__03__min	mfcc__10__skew	mfcc__18__min	mfcc__d2__11__mean
mfcc__03__max	mfcc__10__kurt	mfcc__18__max	mfcc__d2__11__std
mfcc__03__median	mfcc__11__mean	mfcc__18__median	mfcc__d1__12__mean
mfcc__03__iqr	mfcc__11__std	mfcc__18__iqr	mfcc__d1__12__std
mfcc__03__skew	mfcc__11__min	mfcc__18__skew	mfcc__d2__12__mean
mfcc__03__kurt	mfcc__11__max	mfcc__18__kurt	mfcc__d2__12__std
mfcc__04__mean	mfcc__11__median	mfcc__19__mean	mfcc__d1__13__mean
mfcc__04__std	mfcc__11__iqr	mfcc__19__std	mfcc__d1__13__std
mfcc__04__min	mfcc__11__skew	mfcc__19__min	mfcc__d2__13__mean
mfcc__04__max	mfcc__11__kurt	mfcc__19__max	mfcc__d2__13__std
mfcc__04__median	mfcc__12__mean	mfcc__19__median	mfcc__d1__14__mean
mfcc__04__iqr	mfcc__12__std	mfcc__19__iqr	mfcc__d1__14__std
mfcc__04__skew	mfcc__12__min	mfcc__19__skew	mfcc__d2__14__mean
mfcc__04__kurt	mfcc__12__max	mfcc__19__kurt	mfcc__d2__14__std
mfcc__05__mean	mfcc__12__median	mfcc__d1__00__mean	mfcc__d1__15__mean
mfcc__05__std	mfcc__12__iqr	mfcc__d1__00__std	mfcc__d1__15__std
mfcc__05__min	mfcc__12__skew	mfcc__d2__00__mean	mfcc__d2__15__mean
mfcc__05__max	mfcc__12__kurt	mfcc__d2__00__std	mfcc__d2__15__std
mfcc__05__median	mfcc__13__mean	mfcc__d1__01__mean	mfcc__d1__16__mean
mfcc__05__iqr	mfcc__13__std	mfcc__d1__01__std	mfcc__d1__16__std
mfcc__05__skew	mfcc__13__min	mfcc__d2__01__mean	mfcc__d2__16__mean
mfcc__05__kurt	mfcc__13__max	mfcc__d2__01__std	mfcc__d2__16__std
mfcc__06__mean	mfcc__13__median	mfcc__d1__02__mean	mfcc__d1__17__mean
mfcc__06__std	mfcc__13__iqr	mfcc__d1__02__std	mfcc__d1__17__std
mfcc__06__min	mfcc__13__skew	mfcc__d2__02__mean	mfcc__d2__17__mean
mfcc__06__max	mfcc__13__kurt	mfcc__d2__02__std	mfcc__d2__17__std
mfcc__06__median	mfcc__14__mean	mfcc__d1__03__mean	mfcc__d1__18__mean
mfcc__06__iqr	mfcc__14__std	mfcc__d1__03__std	mfcc__d1__18__std
mfcc__06__skew	mfcc__14__min	mfcc__d2__03__mean	mfcc__d2__18__mean
mfcc__06__kurt	mfcc__14__max	mfcc__d2__03__std	mfcc__d2__18__std
mfcc__07__mean	mfcc__14__median	mfcc__d1__04__mean	mfcc__d1__19__mean
mfcc__07__std	mfcc__14__iqr	mfcc__d1__04__std	mfcc__d1__19__std
mfcc__07__min	mfcc__14__skew	mfcc__d2__04__mean	mfcc__d2__19__mean
mfcc__07__max	mfcc__14__kurt	mfcc__d2__04__std	mfcc__d2__19__std

TABLE XVII
Prosodic and voice-quality descriptors (Praat): all 56 column names.

prosody_f0_mean	prosody_jitter_local	prosody_hnr_median	prosody_f4_std
prosody_f0_std	prosody_jitter_rap	prosody_f1_mean	prosody_f4_median
prosody_f0_min	prosody_jitter_ppq5	prosody_f1_std	prosody_f5_mean
prosody_f0_max	prosody_jitter_ddp	prosody_f1_median	prosody_f5_std
prosody_f0_median	prosody_shimmer_local	prosody_f1_bw_median	prosody_f5_median
prosody_f0_iqr	prosody_shimmer_localdb	prosody_f2_mean	prosody_intensity_mean
prosody_f0_skew	prosody_shimmer_apq3	prosody_f2_std	prosody_intensity_std
prosody_f0_kurt	prosody_shimmer_apq5	prosody_f2_median	prosody_intensity_min
prosody_voiced_fraction	prosody_shimmer_apq11	prosody_f2_bw_median	prosody_intensity_max
prosody_f0_slope_abs_mean	prosody_shimmer_dda	prosody_f3_mean	prosody_intensity_median
prosody_n_voiced_segments	prosody_hnr_mean	prosody_f3_std	prosody_intensity_iqr
prosody_voiced_rate_per_s	prosody_hnr_std	prosody_f3_median	prosody_intensity_skew
prosody_mean_voiced_seg_s	prosody_hnr_min	prosody_f3_bw_median	prosody_intensity_kurt
prosody_pause_fraction	prosody_hnr_max	prosody_f4_mean	prosody_cpps

TABLE XVIII
Chroma descriptors: all 24 column names.

chroma__00__mean	chroma__04__mean	chroma__08__mean
chroma__00__std	chroma__04__std	chroma__08__std
chroma__01__mean	chroma__05__mean	chroma__09__mean
chroma__01__std	chroma__05__std	chroma__09__std
chroma__02__mean	chroma__06__mean	chroma__10__mean
chroma__02__std	chroma__06__std	chroma__10__std
chroma__03__mean	chroma__07__mean	chroma__11__mean
chroma__03__std	chroma__07__std	chroma__11__std

TABLE XIX
Global descriptors: all 2 column names.

duration__s	peak__amplitude

TABLE XX
Depth-3 tree fitted on Uncompressed WAV.

Path	Feature	Threshold	H	IG	n
root	prosody__intensity__std	8.2184	2.5850	0.3090	4905
L	mfcc__d1__00__std	3.9965	2.3849	0.1415	3065
LL	spectral__band__0__500__mean	0.8626	1.9382	0.0685	1059
LR	prosody__f0__mean	220.5886	2.3995	0.0990	2006
R	mfcc__d1__00__std	6.6630	2.0891	0.1797	1840
RL	prosody__f0__median	161.8673	2.2633	0.1141	847
RR	prosody__intensity__min	42.7554	1.6055	0.0925	993

TABLE XXI
Depth-3 tree fitted on MP3 @ 64 kbit/s.

Path	Feature	Threshold	H	IG	n
root	prosody__intensity__std	8.2210	2.5850	0.3090	4905
L	mfcc__d1__00__std	4.0175	2.3849	0.1398	3065
LL	spectral__band__0__500__mean	0.8627	1.9524	0.0669	1080
LR	prosody__f0__mean	221.5304	2.3997	0.1002	1985
R	mfcc__d1__00__std	6.6883	2.0891	0.1800	1840
RL	prosody__f0__median	161.8687	2.2615	0.1129	873
RR	prosody__intensity__min	42.3291	1.5890	0.0914	967

TABLE XXII
Depth-3 tree fitted on MP4/AAC @ 64 kbit/s.

Path	Feature	Threshold	H	IG	n
root	prosody__intensity__std	8.3241	2.5850	0.3064	4905
L	mfcc__00__iqr	60.5853	2.3887	0.1408	3075
LL	spectral__band__500__1000__mean	0.0797	1.9819	0.0875	1181
LR	prosody__f0__mean	219.0372	2.4096	0.0957	1894
R	mfcc__d1__00__std	7.1056	2.0879	0.1555	1830
RL	prosody__f0__median	161.0991	2.2673	0.1047	874
RR	prosody__intensity__std	11.2496	1.6245	0.0884	956

TABLE XXIII
Depth-3 tree fitted on Roundtrip WAV (AAC re-wrapped).

Path	Feature	Threshold	H	IG	n
root	prosody__intensity__std	8.3241	2.5850	0.3064	4905
L	mfcc__00__iqr	60.5802	2.3887	0.1408	3075
LL	spectral__band__500__1000__mean	0.0782	1.9819	0.0882	1181
LR	prosody__f0__mean	219.0372	2.4096	0.0957	1894
R	mfcc__d1__00__std	7.1058	2.0879	0.1555	1830
RL	prosody__f0__median	161.0993	2.2673	0.1047	874
RR	prosody__intensity__std	11.2497	1.6245	0.0884	956